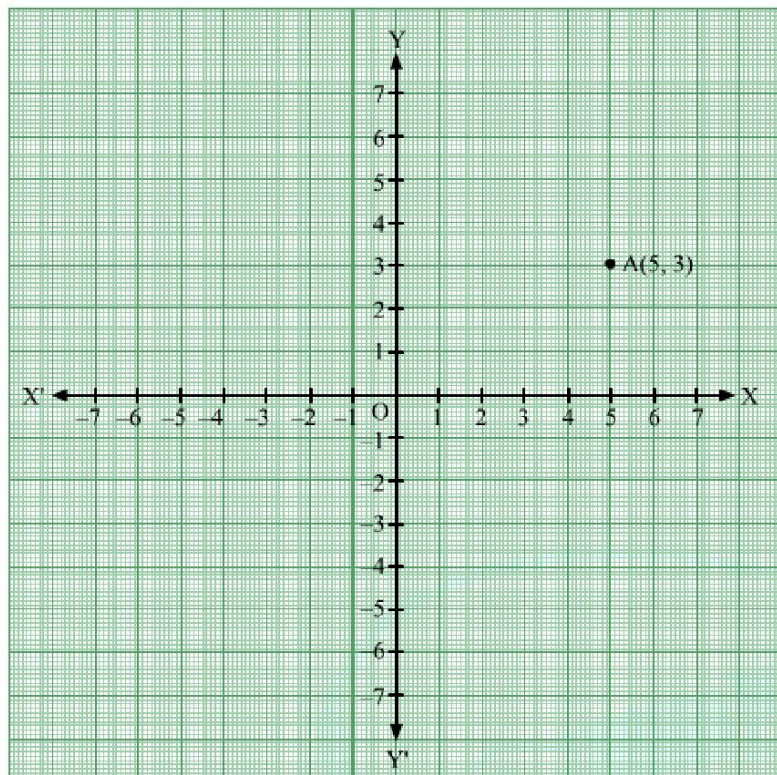


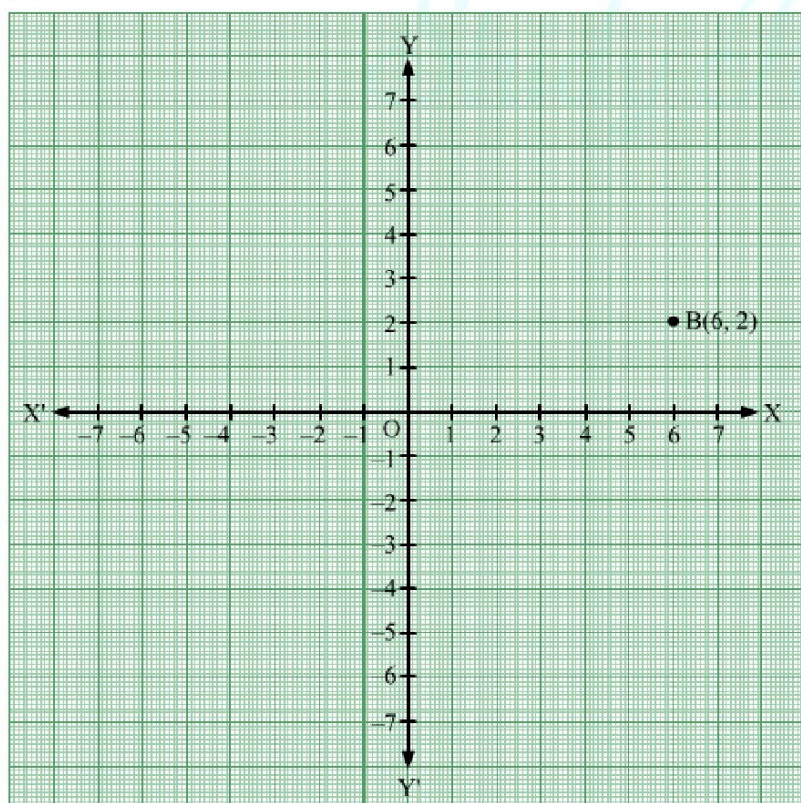
Exercise 5

Solution 1:

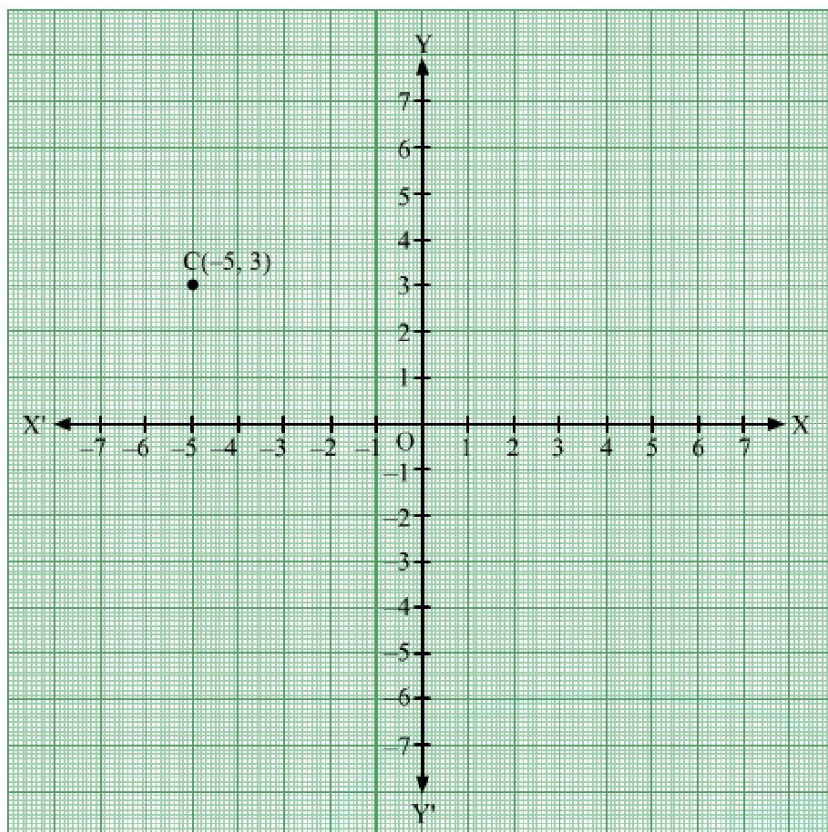
(i)



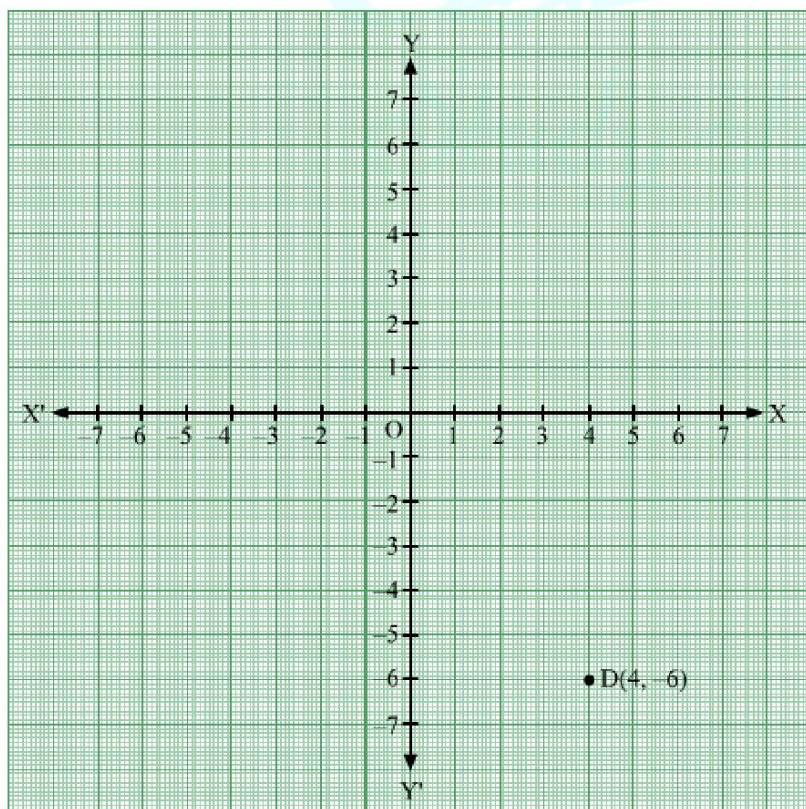
(ii)



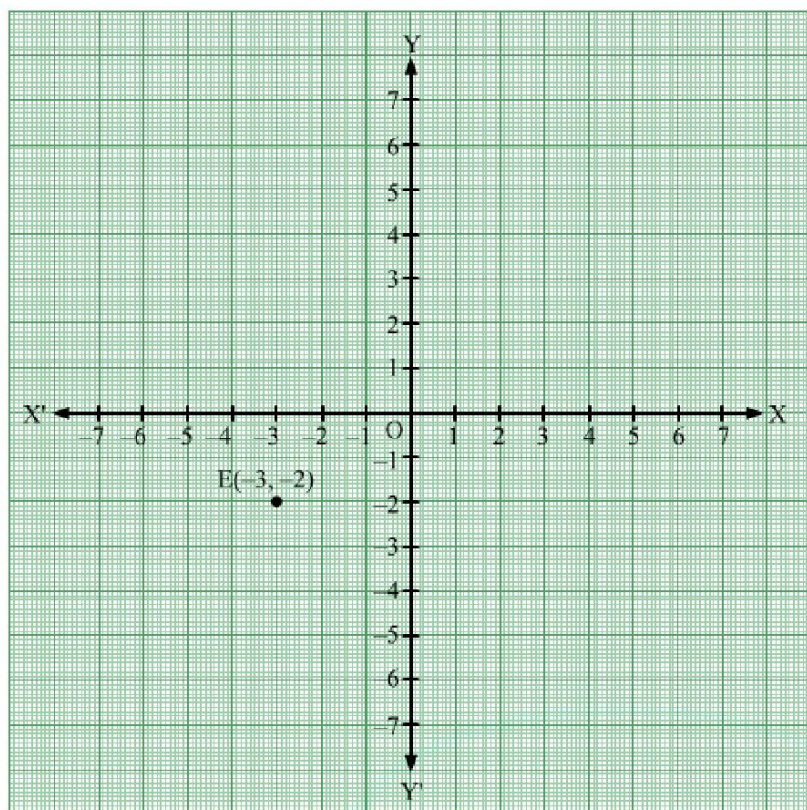
(iii)



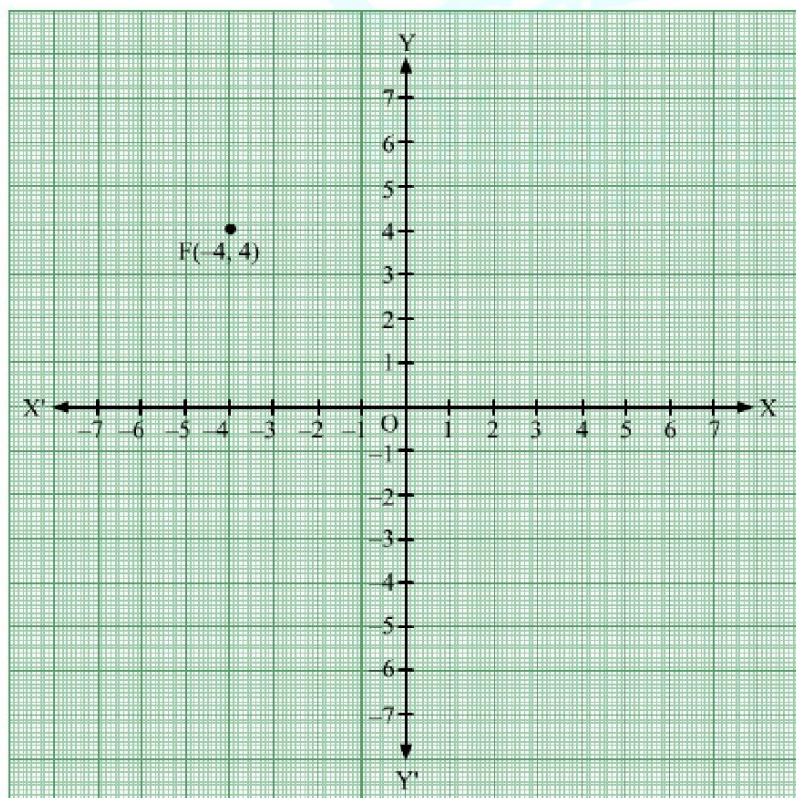
(iv)



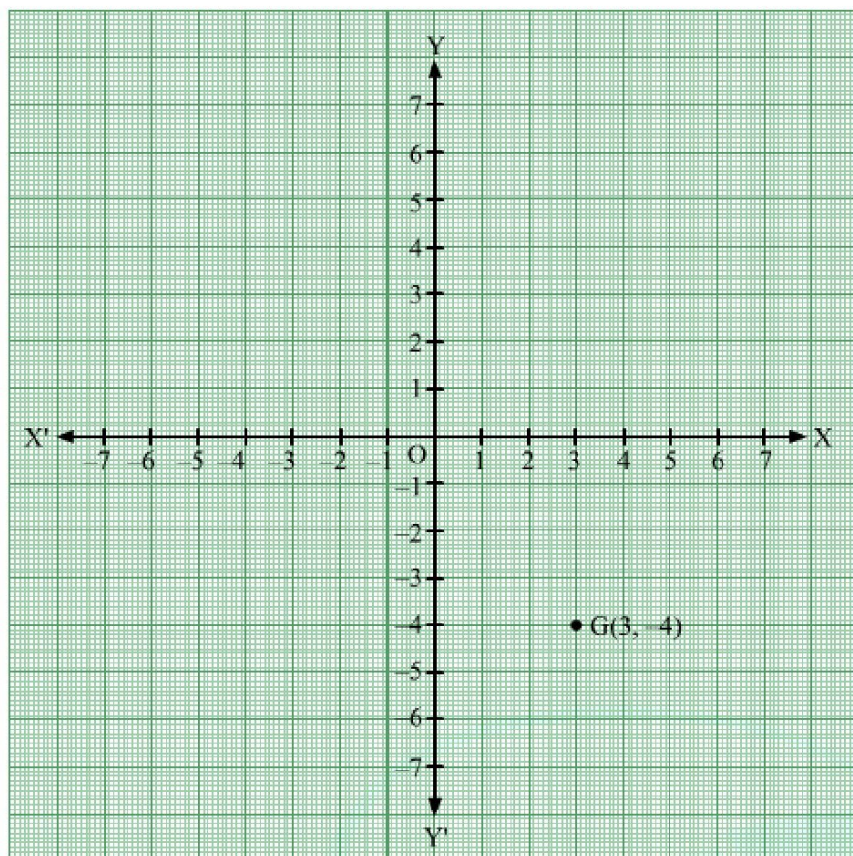
(v)



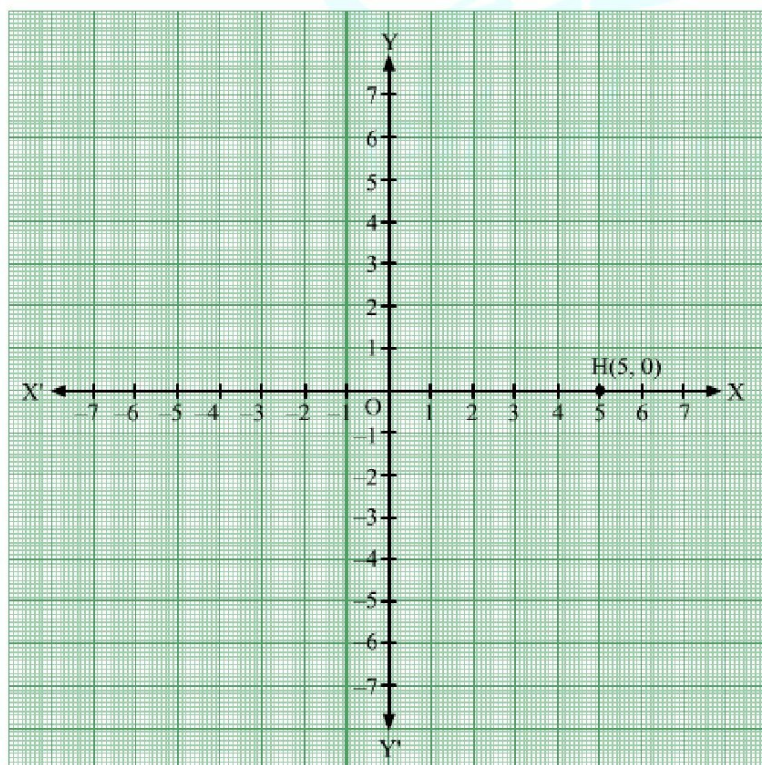
(vi)



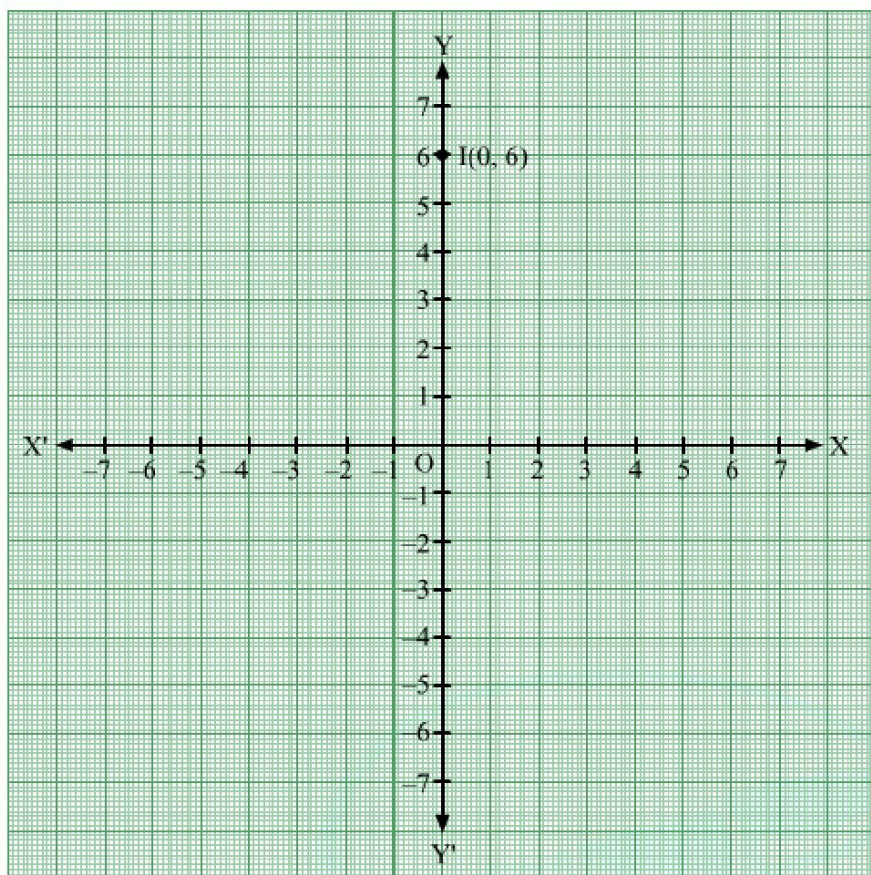
(vii)



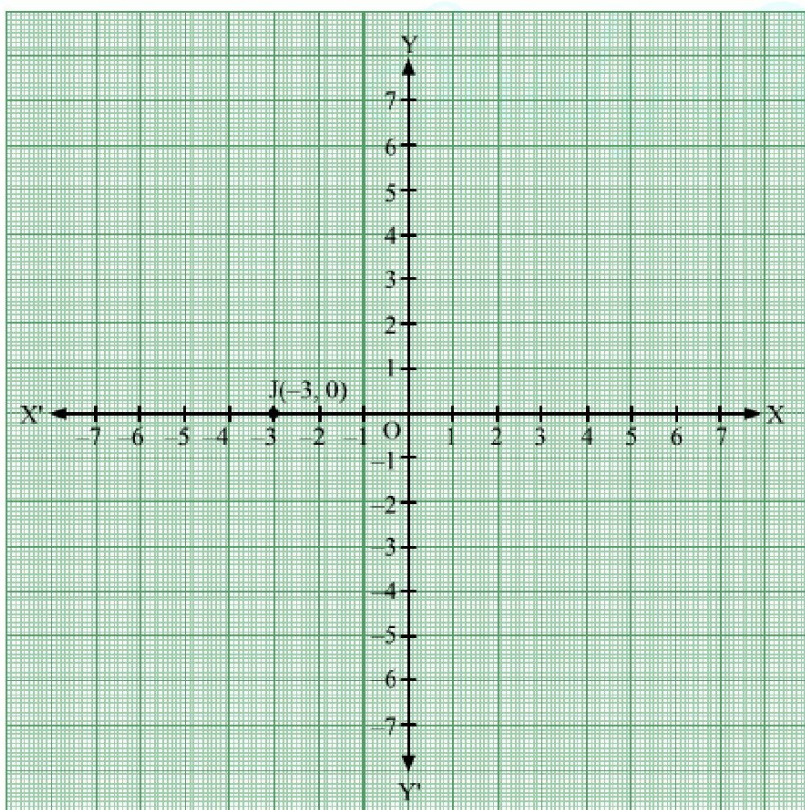
(viii)



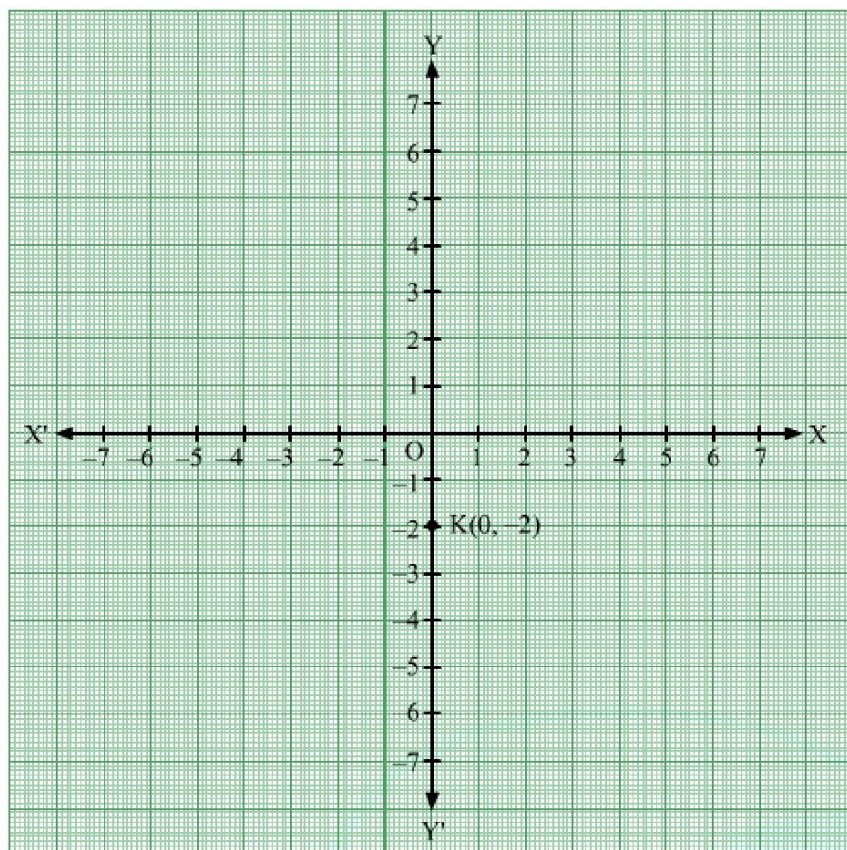
(ix)



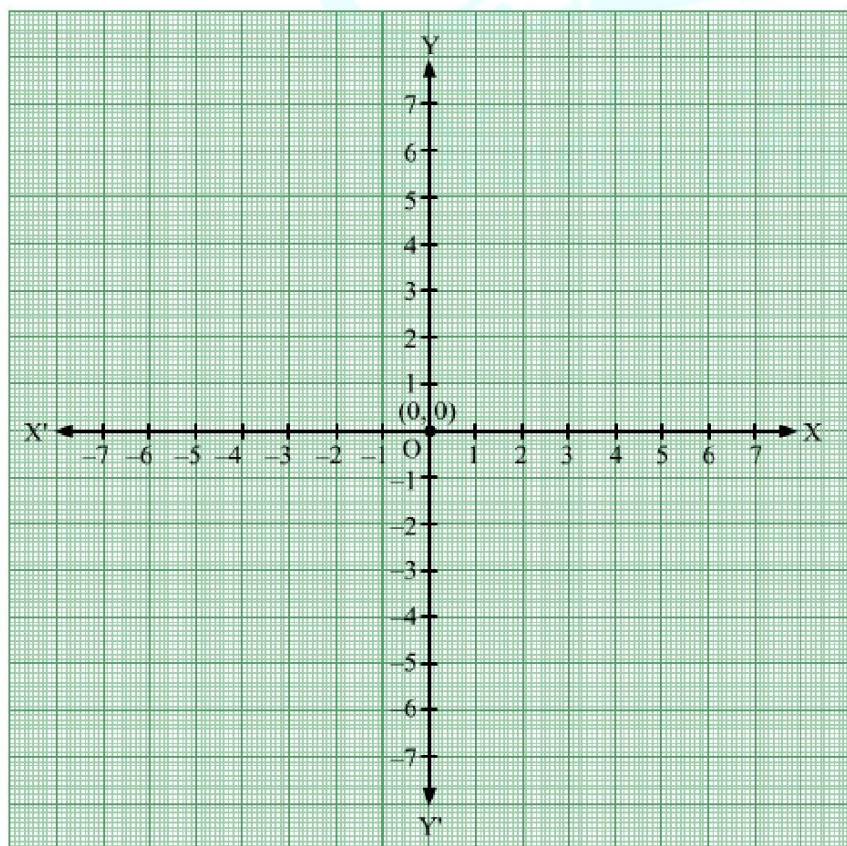
(x)



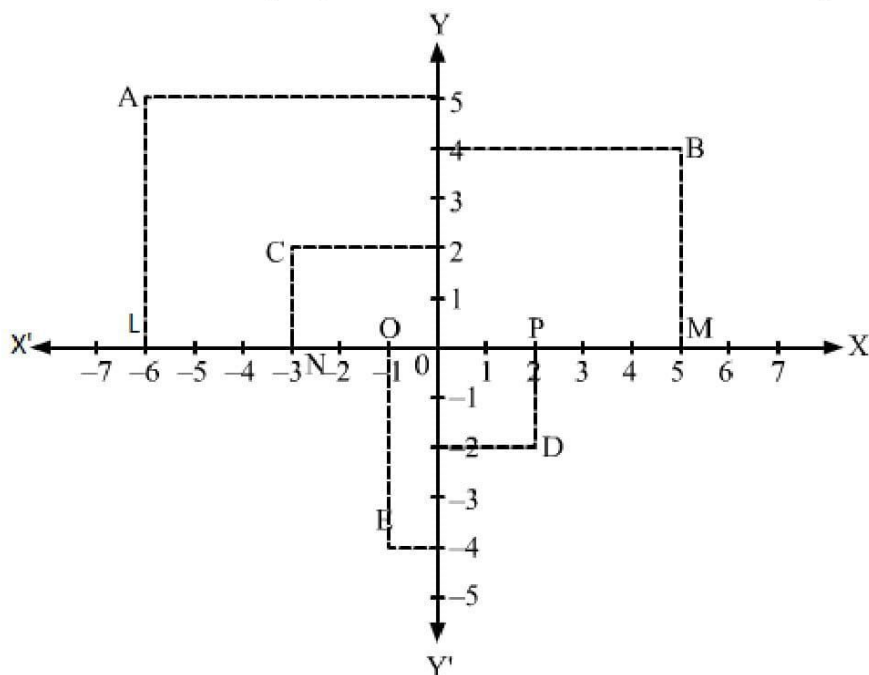
(xi)



(xii)



Solution 2: Draw perpendicular AL, BM, CN, DP and EQ on the X-axis.



(i) Distance of A from the Y-axis = OL = -6 units

Distance of A from the X-axis = AL = 5 units

Hence, the coordinates of A are (-6, 5).

(ii) Distance of B from the Y-axis = OM = 5 units

Distance of B from the X-axis = BM = 4 units

Hence, the coordinates of B are (5, 4).

(iii) Distance of C from the Y-axis = ON = -3 units

Distance of C from the X-axis = CN = 2 units

Hence, the coordinates of C are (-3, 2).

(iv) Distance of D from the Y-axis = OP = 2 units

Distance of D from the X-axis = DP = -2 units

Hence, the coordinates of D are (2, -2).

(v) Distance of E from the Y-axis = OL = -1 units

Distance of E from the X-axis = AL = -4 units

Hence, the coordinates of E are (-1, -4).

Solution 3:(i) $(-6, 3)$ Points of the type $(-, +)$ lie in the II quadrant.Hence, the point lies $(-6, 3)$ in the II quadrant.(ii) $(-5, -3)$ Points of the type $(-, -)$ lie in the III quadrant.Hence, the point lies $(-5, -3)$ in the III quadrant.(iii) $(11, 6)$ Points of the type $(+, +)$ lie in the I quadrant.Hence, the point lies $(11, 6)$ in the I quadrant.(iv) $(1, -4)$ Points of the type $(+, -)$ lie in the IV quadrant.Hence, the point lies $(1, -4)$ in the IV quadrant.(v) $(-7, -4)$ Points of the type $(-, -)$ lie in the III quadrant.Hence, the point lies $(-7, -4)$ in the III quadrant.(vi) $(4, -1)$ Points of the type $(+, -)$ lie in the IV quadrant.Hence, the point lies $(4, -1)$ in the IV quadrant.(vii) $(-3, 8)$ Points of the type $(-, +)$ lie in the II quadrant.Hence, the point lies $(-3, 8)$ in the II quadrant.(viii) $(3, -8)$

Points of the type $(+, -)$ lie in the IV quadrant.

Hence, the point lies $(3, -8)$ in the IV quadrant.

Solution 4:

(i) $(2, 0)$

The ordinate of the point $(2, 0)$ is zero.

Hence, the $(2, 0)$ lies on the x-axis.

(ii) $(0, -5)$

The abscissa of the point $(0, -5)$ is zero.

Hence, the $(0, -5)$ lies on the y-axis.

(iii) $(-4, 0)$

The ordinate of the point $(-4, 0)$ is zero.

Hence, the $(-4, 0)$ lies on the x-axis.

(iv) $(0, -1)$

The abscissa of the point $(0, -1)$ is zero.

Hence, the $(0, -1)$ lies on the y-axis.

Solution 5:

(i) $A(0, 8)$

The given point does not lie on the x-axis.

(ii) $B(4, 0)$

The ordinate of the point $(4, 0)$ is zero.

Hence, the $(4, 0)$ lies on the x-axis.

(iii) $C(0, -3)$

The given point does not lie on the x-axis.

(iv) $D(-6, 0)$

The ordinate of the point $(-6, 0)$ is zero.

Hence, the $(-6, 0)$ lies on the x-axis.

(v) $E(2, 1)$

The given point does not lie on the x-axis.

(vi) $F(-2, -1)$

The given point does not lie on the x-axis.

(vii) $G(-1, 0)$

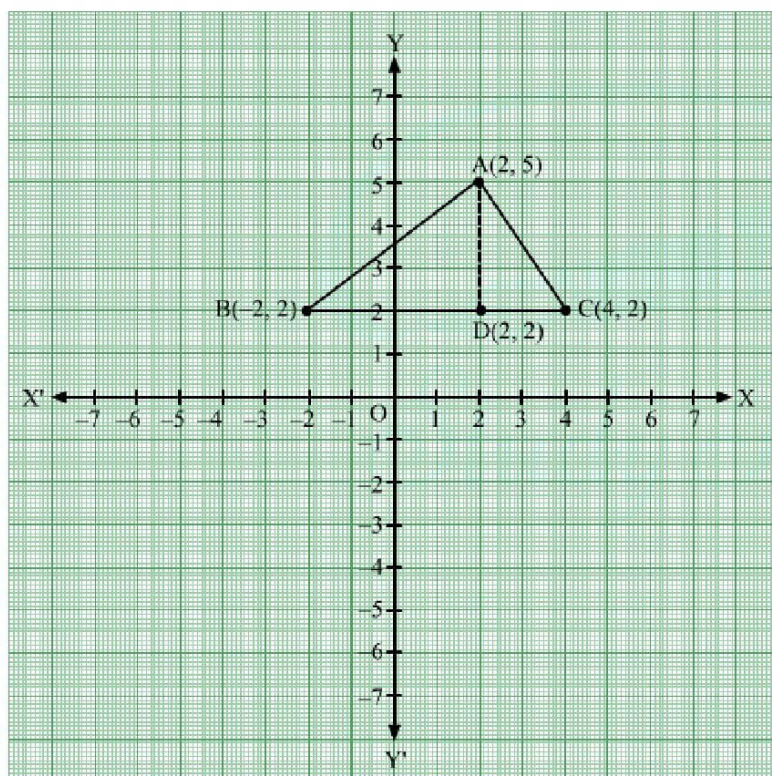
The ordinate of the point $(-1, 0)$ is zero.

Hence, the $(-1, 0)$ lies on the x-axis.

(viii) $H(0, -2)$

The given point does not lie on the x-axis.

Solution 6:



Abscissa of D = Abscissa of A = 2

Ordinate of D = Ordinate of B = 2

Now,

$BC = (2 + 4) \text{ units} = 6 \text{ units}$

$AD = (5 - 2) \text{ units} = 3 \text{ units}$

Area of $\triangle ABC$

$$= \frac{1}{2} \times \text{Base} \times \text{Height}$$

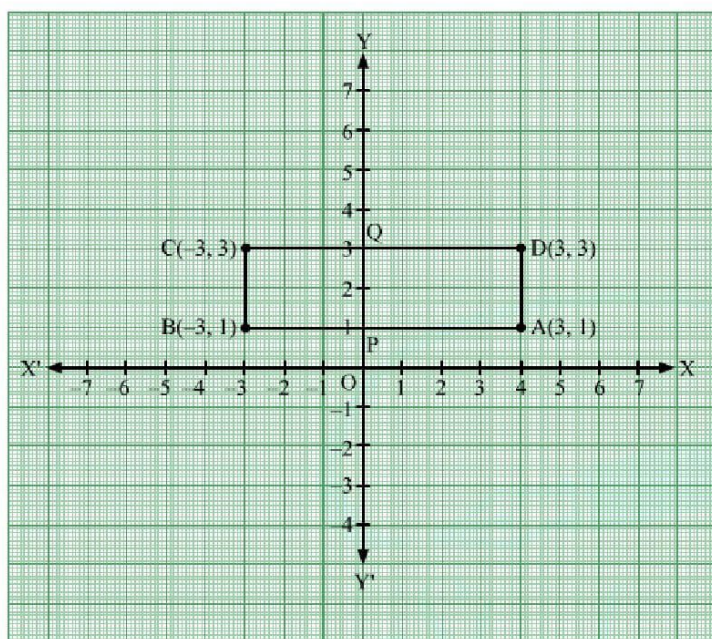
$$= \frac{1}{2} \times BC \times AD$$

$$= \frac{1}{2} \times 6 \times 3$$

$$= 9$$

Hence, area of ΔABC is 9 square units.

Solution 7: Let $A(3, 1)$, $B(-3, 1)$ and $C(-3, 3)$ be three vertices of a rectangle ABCD. Let the y-axis cut the rectangle ABCD at the points P and Q respectively.



Abscissa of D = Abscissa of A = 3.

Ordinate of D = Ordinate of C = 3.

$\therefore$ coordinates of D are (3, 3).

$$AB = (BP + PA) = (3 + 3) \text{ units} = 6 \text{ units.}$$

$$BC = (OQ - OP) = (3 - 1) \text{ units} = 2 \text{ units.}$$

$$\begin{aligned} \text{Ar}(\text{rectangle ABCD}) &= (AB \times BC) \\ &= (6 \times 2) \text{ sq. units} \\ &= 12 \text{ sq. units} \end{aligned}$$

Hence, the area of rectangle ABCD is 12 square units.

Solution 1: (c) III

Points of the type $(-, -)$ lie in the III quadrant.

The point $(-7, -4)$ lies in the III quadrant.

Hence, the correct option is (c).

Solution 2: (d) IV

Explanation:

The points of the type $(+,-)$ lie in fourth quadrant.

Hence, the point (x, y) , where $x > 0$ and $y < 0$, lies in quadrant IV.

Solution 3: (b) II

Explanation:

Points of the type $(-,+)$ lie in the second quadrant.

Hence, the point $P(a,b)$, where $a < 0$ and $b > 0$, lie in quadrant II.

Solution 4: (c) quadrant III

Explanation:

Points of the type $(-,-)$ lie in the third quadrant.

Solution 5: (c) I or III

Explanation:

If abscissa = ordinate, there could be two possibilities.

Either both are positive or both are negative. So, a point could be either $(+,+)$, which lie in quadrant I or it could be of the type $(-,-)$, which lie in quadrant III.

Hence, the points (other than the origin) for which the abscissas are equal to the ordinates lie in quadrant I or III.

Solution 6: (c) II and IV quadrants respectively

The point $(-5, 3)$ lies in the II quadrant.
The point $(3, -5)$ lies in the IV quadrant.
Hence, the correct option is (c).

Solution 7: (d) do not lie in the same quadrant

The point $(1, -1)$ lies in the IV quadrant.
The point $(2, -2)$ lies in the IV quadrant.
The point $(-3, -4)$ lies in the III quadrant.
The point $(4, -5)$ lies in the IV quadrant.

Hence, the correct option is (d).

Solution 8: (d) on the y-axis

The abscissa of the point $(0, -8)$ is zero.
The point $(0, -8)$ lies on the y-axis.
Hence, the correct option is (d).

Solution 9: (a) on the negative direction of the x-axis

The point $(-7, 0)$ lies on the negative direction of the x-axis.
Hence, the correct option is (a).

Solution 10: (b) $(0, -5)$

The point which lies on the y-axis at a distance of 5 units in the negative direction of the y-axis is $(0, -5)$.
Hence, the correct option is (b).

Solution 11: (c) 0

The ordinate of every point on the x-axis is 0.
Hence, the correct option is (c).

Solution 12: (b) on the x-axis

The coordinates of a point on the x-axis are of the form $(x, 0)$ and that of the point on the y-axis is of the form $(0, y)$.

Thus, if the y-coordinate of a point is zero, then this point always lies on the x-axis.

Hence, the correct answer is option (b).

Solution 13: (b) rectangle

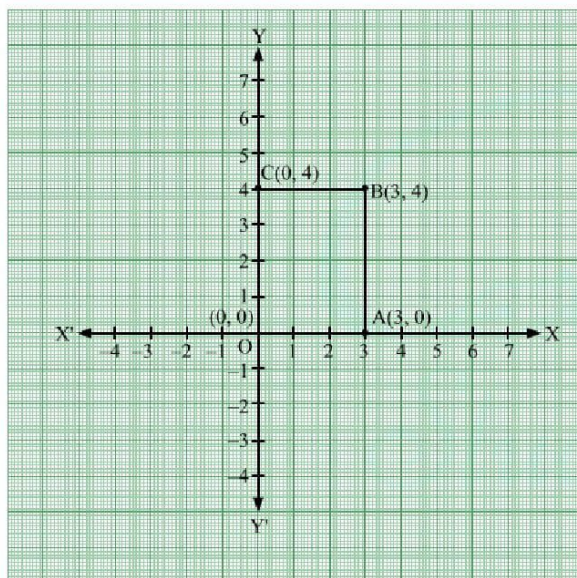
The point $O(0, 0)$ is the origin.

$A(3, 0)$ lies on the positive direction of x-axis.

$B(3, 4)$ lies in the Ist quadrant.

$C(0, 4)$ lies on the positive direction of y-axis.

The points $O(0, 0)$, $A(3, 0)$, $B(3, 4)$ and $C(0, 4)$ can be plotted on the Cartesian plane as follows:



Here, the figure $OABC$ is a rectangle.

Hence, the correct answer is option (b).

Solution 14: (b) 1

The given points are $A(-2, 3)$ and $B(-3, 5)$.

Abscissa of $A = x$ -coordinate of $A = -2$

Abscissa of $B = x$ -coordinate of $B = -3$

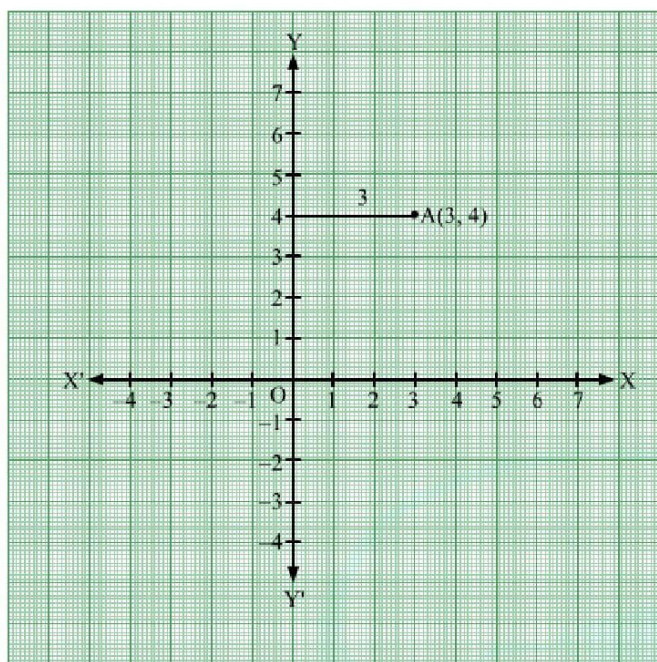
$\therefore$ Abscissa of $A - \text{Abscissa of } B = -2 - (-3) = -2 + 3 = 1$

Hence, the correct answer is option (b).

Solution 15: (a) 3

The perpendicular distance of a point from the y-axis is equal to the x-coordinate of the point.

$\therefore$ Perpendicular distance of the point A(3, 4) from the y-axis = x-coordinate of A(3, 4)
= 3



Hence, the correct answer is option (a).

Solution 16: (b) I and IV quadrants

Explanation:

If abscissa of a point is positive, then the ordinate could be either positive or negative.

It means that the type of any point can be either (+, +) or (+, -).

Points of the type (+, +) lie in quadrant I, whereas points of the type (+, -) lie in quadrant IV.

Solution 17: (c) the origin

Explanation: The point at which two axes meet is called as the origin.

Solution 18: (b) (0, 3)

The ordinate of a point is the y-coordinate of the point. So, the y-coordinate of the point is 3.

Also, any point on the y-axis has coordinates in the form $(0, y)$.

Thus, the point whose ordinate is 3 and which lies on the y-axis is $(0, 3)$.

Hence, the correct answer is option (b).

Solution 19: (b) $(3, 9)$

Explanation:

Point $(2, 8)$ does not satisfy the equation $y = 2x + 3$. ($\because y = 2 \times 2 + 3 = 7 \neq 8$)

Point $(3, 9)$ satisfy the equation $y = 2x + 3$. ($\because y = 2 \times 3 + 3 = 9$)

Point $(4, 12)$ does not satisfy the equation $y = 2x + 3$. ($\because y = 2 \times 4 + 3 = 11 \neq 12$)

Point $(5, 15)$ does not satisfy the equation $y = 2x + 3$. ($\because y = 2 \times 5 + 3 = 13 \neq 15$)

Hence, the point $(3, 9)$ lies on the line $y = 2x + 3$.

Solution 20: (d) $(4, 12)$

Explanation:

(a) Point $(1, 7)$ satisfy the equation $y = 3x + 4$. ($\because y = 3 \times 1 + 4 = 7$)

(b) Point $(2, 10)$ satisfy the equation $y = 3x + 4$. ($\because y = 3 \times 2 + 4 = 10$)

(c) Point $(-1, 1)$ satisfy the equation $y = 3x + 4$. ($\because y = 3 \times -1 + 4 = 1$)

(d) Point $(4, 12)$ does not satisfy the equation $y = 3x + 4$. ($\because y = 3 \times 4 + 4 = 16 \neq 12$)

Hence, the point $(4, 12)$ do not lie on the line $y = 3x + 4$.

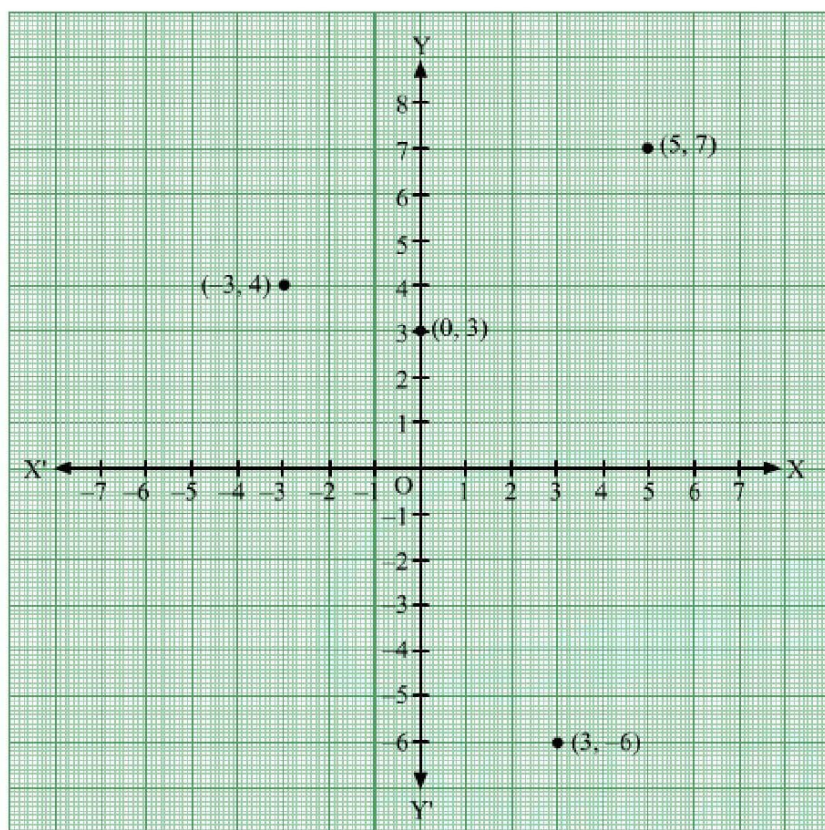
Solution 21: (d) $(0, 3)$

The point $(3, -6)$ lies in the fourth quadrant.

The point $(-3, 4)$ lies in the second quadrant.

The point $(5, 7)$ lies in the first quadrant.

The point $(0, 3)$ lies on the positive direction of y-axis.

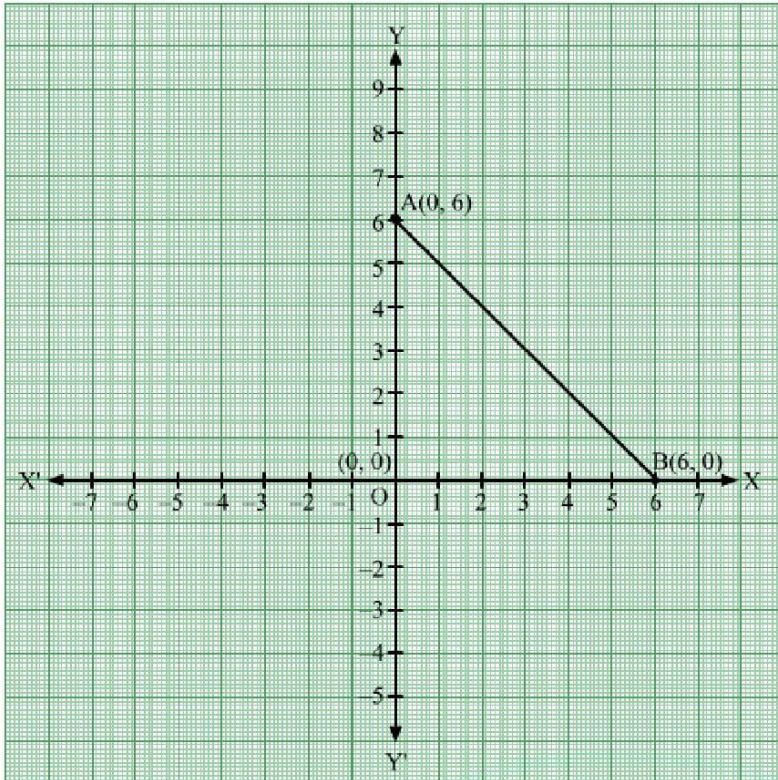


Thus, the point $(0, 3)$ does not lie in any quadrant.

Hence, the correct answer is option (d).

Solution 22: (c) 18 sq units

The points $A(0, 6)$, $O(0, 0)$ and $B(6, 0)$ can be plotted on the Cartesian plane as follows:



Here, $\triangle AOB$ is a right triangle right angled at O.

$OA = 6$ units and $OB = 6$ units

$$\therefore \text{Area of } \triangle AOB = \frac{1}{2} \times OA \times OB$$

$$= \frac{1}{2} \times 6 \times 6$$

$$= 18 \text{ square units}$$

Hence, the correct answer is option (c).